

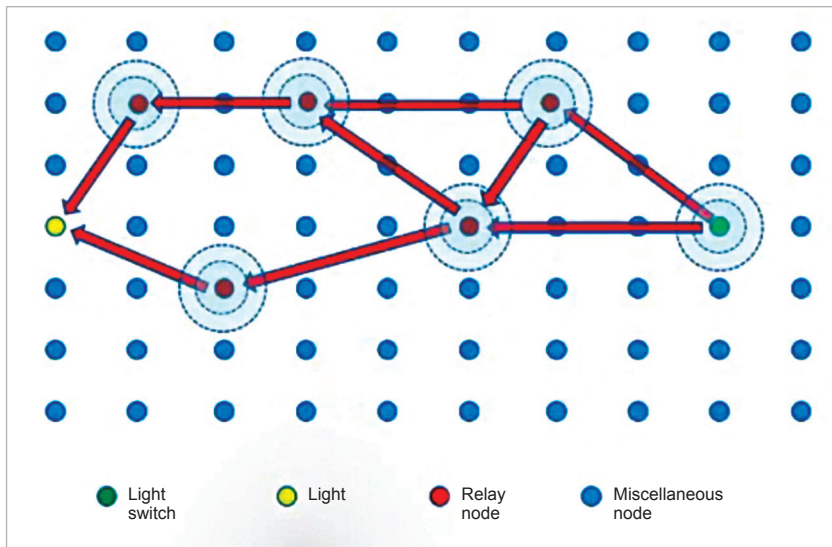


Figure 15: Multiple redundant paths across the network

another form of advertising called extended advertising, which uses all 40 Bluetooth LE channels.

The use of three channels in basic advertising is less sophisticated than adaptive frequency hopping in a connection. But the advertising channels are deliberately placed far apart from each other so that interference in one part does not block communication completely.

Basic advertising happens at defined intervals, but it includes some random timing variation in the scheduling of packet transmission of up to ten milliseconds. This is to avoid persistent collisions with nearby devices that happen to be transmitting or advertising on the same schedule.

Bluetooth connectionless communication does not attempt to coordinate the activities of the advertising device with that of the receiving devices. However, there is a special advertising mode called periodic advertising, which makes the timing of the advertising completely deterministic and precise.

And scanning devices can also discover the periodic advertising schedule of the broadcasting device so that they can synchronise their scanning activity precisely with it. This

way, devices can more reliably and efficiently receive broadcast data than with the basic form of advertising.

Bluetooth mesh networks

Bluetooth mesh allows creation of large networks, having tens and thousands of devices. It is a special way of using Bluetooth LE where the mesh stack sits on top of the Bluetooth LE controller.

Bluetooth mesh defines bearers, which provide different ways of using the Bluetooth LE controller for the communication of mesh messages. And the bearers provide support for both connectionless and connection-oriented communication.

Addressing Bluetooth mesh issues

Mesh nodes can be configured to automatically retransmit messages at defined intervals, so that multiple copies get rapidly retransmitted.

And, consequently, the probability of message loss drops dramatically when you send multiple copies. If you want to ensure that messages are acted upon by all receiving devices at the same time (even if they do not receive the same copy), that can be done too by exercising a delay parameter, which is available in many of the mesh message types.

Figure 12 shows message being transmitted three times at 50ms intervals. Here, the first copy has a delay parameter of 100ms and is received by four of the six target lights.

Figure 13 shows the second copy is sent 50ms later and is received by the fifth of the six lights.

And then the final copy has no delay specified (Figure 14). It is received by the second of the six lights. Then all six nodes execute the requested state change in unison, exactly 100ms after the light switch was pressed. This avoids what is known as the popcorn effect, where lights do not come on at the same time.

Bluetooth mesh-multipath delivery

A common network design strategy for Bluetooth mesh is to build multiple redundant paths (Figure 15) across the network by thoughtfully enabling the relay function in a subset of your nodes. Then, when you send a message, all of these paths are available in principle. In practice, if one path is unavailable due to the duty cycle of a node, there is a probability that one of the other paths will be available and your transmitted message will get to its destination at the end of the path. **END** 🐼

The article is based on the talk 'Understanding reliability in Bluetooth technology' by Martin Woolley, senior developer relations manager, EMEA at Bluetooth SIG at February edition of Tech World Congress and India Electronics Week 2021. A comprehensive paper on the subject, written by Martin, is available from the Bluetooth SIG website at <https://www.bluetooth.com/bluetooth-resources/understanding-reliability-in-bluetooth-technology/>

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